

Anton Petnitsky

Full-Stack Developer · Python & TypeScript · AI/LLM Integration · Backend Systems

anton.petnitsky@gmail.com github.com/Mukller linkedin.com/in/anton-petnitsky antonpetnitsky.com

Minsk, Belarus · Remote Only Russian & Belarusian: Native · English: B1–B2

PROFESSIONAL SUMMARY

Full-stack developer with 2+ years of hands-on experience shipping and operating 39 public and 17 private repositories, including production systems used by real businesses and end users. Built and deployed a commercial delivery platform (Telegram bots + Flutter courier app + FastAPI backend + geospatial routing, live for a business in Uzbekistan) and a self-hosted AI assistant with local LLM inference, voice cloning, and a computer-vision video pipeline. Comfortable across the stack — Python/FastAPI backends, Node.js/NestJS services, Flutter mobile, and embedded C++/ESP32 robotics — with a strong focus on database design, real-time systems, and taking a project from idea through Docker-based deployment to live users. Separately, two years of paid academic writing — coursework, diploma projects and theses across engineering and computer science: researching an unfamiliar subject, running the experiment or calculation, and carrying the write-up through to a defended result. Currently a first-year Robotic Systems student at BNTU, Faculty of IT & Robotics, group 10706126.

CORE SKILLS

Languages	Python (primary), TypeScript/JavaScript (Node.js), Dart (Flutter), Java, C++
Backend & APIs	FastAPI, REST APIs, async/await, aiogram 3, Telegraf, NestJS, WebSockets, MTProto (GramJS)
AI / LLM	Generative AI & LLM integration (Claude API, local via Ollama), prompt engineering, LoRA fine-tuning, voice cloning (Coqui XTTS-v2), ONNX Runtime, OpenCV, Stockfish
Databases	PostgreSQL, MySQL, Redis, Prisma ORM, BullMQ job queues
DevOps / Infra	Docker & Compose, CI/CD (GitHub Actions: build, test, automated releases), systemd, Linux (Ubuntu 24.04), Git/GitHub, Tailscale, self-hosted deployment
Mobile	Flutter/Dart, Android release builds & signing
Robotics / Embedded	Arduino, ESP32, ROS 2, PID control, sensor fusion, motor control (ESC/BLDC/L298)
Architecture	Microservices, event-driven design, real-time systems, geospatial routing (OSRM)

PROFESSIONAL EXPERIENCE

Full-Stack Developer — Independent & Freelance

June 2024 – Present

Remote

Designed, built, deployed, and maintain 39 public and 17 private repositories end-to-end. 8+ applications actively serving real users; several run 24/7 in production on self-hosted Linux infrastructure.

Freelance Academic Writer

2 years

Remote

Two years of paid academic work — coursework, diploma projects and final theses in engineering and computer-science subjects. Full cycle: scoping the task, learning material I did not know, carrying out the calculation or experiment,

formatting to the university's requirements, and supporting the student through the defence. It built two skills an ordinary developer does not have: holding the structure of someone else's field, and finishing to formal requirements rather than to "it works on my machine".

Safalife — Water Delivery Platform (Freelance / Commercial) · Private repos

Full-stack delivery platform for a live business in Uzbekistan: client & courier Telegram bots, Flutter courier/dispatcher app, FastAPI backend. HMAC-secured GPS order tracking, role-based access, self-hosted OSRM routing across two-country road network. Deployed via Docker Compose with Tailscale-secured API.

Python · FastAPI · aiogram 3 · Flutter/Dart · PostgreSQL · OSRM · Docker Compose · Tailscale

AI Persona Behavioral OS · github.com/Mukller/ai-persona-behavioral-os

Self-hosted AI assistant platform: FastAPI + PostgreSQL backend, Telegram bot interface, token-authenticated analytics dashboard. Local LLM inference (Ollama, hot-swappable models), zero-shot voice cloning (Coqui XTTS-v2), custom LoRA fine-tuning pipeline trained on ~7,600 conversation pairs. Computer-vision video pipeline generating personalized Telegram video messages via background worker + ONNX Runtime. 24/7 under systemd.

Python · FastAPI · PostgreSQL · WebSockets · Ollama · ONNX Runtime · OpenCV · Docker · systemd · pytest

Chess Bot (Telegram) · github.com/Mukller/chess

Full-featured Telegram chess platform with 3 game modes (AI, hot-seat, PvP). Stockfish engine integration, dynamic difficulty, ELO rating system. 100+ daily users, 1,000+ games played.

Python 3.10+ · aiogram 3 · Stockfish · PostgreSQL · AsyncIO

Telegram Birthday Assistant · Private repo

Userbot that reminds about birthdays and sends congratulations on the user's behalf via MTPROTO (GramJS). 65-template greeting library, Telegram Stars gift sending. Live with 600+ contacts migrated; timezone auto-detected, reminders via Redis/BullMQ job queue.

NestJS 10 · Prisma 5 · PostgreSQL · Redis · BullMQ · Telegraf · GramJS · Docker Compose

ADDITIONAL PROJECTS

- **MiniBoardGames 3.0.0** — 20+ mini-games in one Minecraft plugin (2048, Flappy Bird, Battleship), multiplayer 4–100+ players, persistent leaderboards. *Java, PaperMC, MySQL*
- **Casino Stars Bot** — Telegram slot-machine game monetized via Paid Messages; automated jackpot detection and gift payouts via Telegram Gift API. *Python, aiogram 3.13*
- **Telegram Presence Tracker** — Online/offline monitoring (aiogram bot + Telethon worker), 1s polling with flood-wait backoff. 24/7 on systemd. *Python, PostgreSQL, Redis*
- **TikTok Saver Bot** — Telegram bot for TikTok video/photo/audio downloads; parallel CDN downloads + yt-dlp fallback. 24/7 under systemd. *Python, aiogram 3, yt-dlp*
- **3D Printer Control** — Raspberry Pi stack (OctoPrint + Klipper) + custom Telegram bot, AI print-quality monitoring, smart-plug automation. *Python, OctoPrint*
- **Claude Auto-Continue** — Windows desktop automation (UIA tree traversal) to auto-recover coding-assistant sessions; 3 public releases (v1.0.0–v3.1.0). *Python, Tkinter, UI Automation*
- **AdvancedSpyInventory / HackedServer / DailyStreak** — Minecraft plugins for anti-cheat, inventory monitoring, and activity tracking. *Java, PaperMC, Packet API*
- **Minecraft Server Infrastructure ("Техношилд" / MineLore)** — Velocity proxy + ~150 concurrent players + 10+ plugins + Discord bridge + Bedrock cross-play. *Java, Spigot, Velocity, Geyser*

ROBOTICS & HARDWARE

Arkanoid Robot "Robo Arkasha" — Robofinist 2026

Autonomous robot playing Arkanoid in real-time. PC-side OpenCV pipeline (HSV tracking, Kalman filter) sends ball position over UDP to ESP32 motor controller (BLDC via ESC + dual DC motors).

Python · OpenCV · ESP32 · C++

FIRA Autonomous RC Car · github.com/Mukller/fira-autonomous-rc-car

1:10-scale autonomous RC car for FIRA/RoboCup Autonomous Cars League. ROS 2 package (lane detection, steering/speed control, serial bridge), camera calibration tooling.

ROS 2 · Python · Arduino · C++

Self-Balancing Robot · github.com/Mukller/Balance_robot

Two-wheeled inverted pendulum (ESP32 + MPU6050 + 2× TMC2209 + NEMA17). Custom PID tuning from scratch. Multiple regional/national competition victories; presented at scientific research conferences. 3 GitHub releases.

C++ · Arduino · ESP32

COMPETITIONS & ACHIEVEMENTS

- JuniorSkills Belarus — 1st Place, "Line Following" — team CyberLab (Minsk), March 2023
- Educational Robotics Cup — regional Robofinist-circuit competition, Minsk Region
- XII Minsk Open Robot Tournament — multi-day robotics competition
- XV Frunzensky District Championship — represented Gymnasium №39, first round

EDUCATION

Belarusian National Technical University (BNTU)

Faculty of IT & Robotics (ФИТР) · Robotic Systems · 1st-year student
2026 – Present

Gymnasium №39, Minsk — IT-specialized track

Programming school & IT circle. Winner of robotics competitions, programming contests, and scientific conferences.
2015 – 2026

CURRENTLY LEARNING

Go (Golang), Kubernetes, gRPC, Helm, MQTT, GitLab CI/CD, cloud platforms (AWS/Azure/GCP), Django/Flask, SCADA/PLC/HMI fundamentals.

PORTFOLIO STATS

83

Repositories

8+

Live deployments

100+

Active users

9

Languages

SOFT SKILLS

Direct, honest communication with clients and teammates — clear about scope, timelines, and problems as they come up. Comfortable working remotely across time zones and picking up new tools or domains quickly when a project needs it.

AVAILABLE FOR

Remote positions: Full-time · Contract · Freelance | **Timezone:** UTC/EET (flexible) | **Start:** Immediate
Preferred roles: Backend Engineer · Full-Stack Developer · AI/Bot Developer · Junior Software Engineer